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Denis A. Khorkin
Lobachevsky State University of Nizhny Novgorod,
Arzamas Branch of UNN
e-mail: dkh01101111@gmail.com, deneeze@yandex.ru

Technical analysis of volatility

The paper is devoted to the volatility cone as a technical analysis tool for forecasting volatility. A volatility cone is constructed, factors and basic properties are identified, the relationship of different types of volatility is shown, and the basic principle of volatility analysis is formulated.

Keywords: *option, volatility, volatility cone, technical analysis*

Introduction

When trading options, every trader has a question of whether options are cheap or expensive. To answer this question, it is necessary to determine the value of the option. There are different options pricing models, such as the Black-Scholes model[1] and its modifications[2] (for European options), Yates model[3] (an improved version of Black-Scholes model which accounts for dividends and possibility of early exercise), Garman-Kohlhagen model[4] (for foreign currency options), Cox-Ross-Rubinstein model[5] (binomial model), quadratic approximation Whaley model[6] (for American options), Bjerksund-Stensland model[7] (for American options), Heston model [8] (uses stochastic volatility for determination of European options price), Monte Carlo model[9] and others.

For definiteness, let us take the Black-Scholes model. This model is fairly simple and does not have a considerable number of input parameters. As an example, let us take a modification of the Black-Scholes model for European futures options free of premium (marginable options). The general equation of the estimation model can be written as follows: [1]:

$$option\ price\ =\ f(futures\ price, strike, time, rate, volatility)$$

Black-Scholes formula for computation of marginable option theoretical value (let us assume that the risk-free interest rate $r\ =0$):

The following notations are used in the formulas below:

C – theoretical value of a call option on futures without paying a premium;

P – theoretical value of a put option on futures without paying a premium;

F – current price of a futures contract;

E – strike price;

T – time to expiration in years;

σ – annual volatility;

e – base of natural logarithm;

ln – natural logarithm;

$\varphi(x)$ – density function of standard normal distribution ($\mu\ =0$ и $\sigma\ =1$)

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}, \quad (1)$$

$N(x)$ – standard normal distribution function

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$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{\frac{-x^2}{2}}, \quad (2)$$

N(d1), N(d2) – probabilities determined from the table of values of normal probability distribution series.

$$C = FN(d1) - EN(d2), \qquad P = C + (E - F), \quad (3)(4)$$

where

$$d1 = \frac{ln\left(\frac{F}{E}\right) + 0.5\sigma^2 T}{\sigma\sqrt{T}}, \qquad d2 = \frac{ln\left(\frac{F}{E}\right) - 0.5\sigma^2 T}{\sigma\sqrt{T}} = d1 - \sigma\sqrt{T}$$

Sensitivity indices (the Greeks) for Black-Scholes formula:

$$\Delta_c = N(d1), \qquad \Delta_p = -N(-d1), \quad (5)(6)$$

$$\Gamma = \frac{\varphi(d1)}{F\sigma\sqrt{T}}, \quad (7)$$

$$\Theta_c = \Theta_p = -\frac{F\sigma\varphi(d1)}{2\sqrt{T}}, \quad (8)$$

$$Vega = F\sqrt{T}\varphi(d1), \quad (9)$$

$$\rho_c = \rho_p = 0, \quad (10)$$

Looking at Black-Scholes formula above, it becomes apparent that all parameters, except volatility, are known quantities. The question then becomes what volatility should be used in the model. There are 4 types of volatility:

1. Future ("true") volatility (nobody can derive it);
2. Historical volatility (computed exactly);
3. Forecast volatility (anticipated or predicted);
4. Implied volatility (Newton-Raphson method makes it possible to quickly obtain this value[10]).

Traders use different volatility indicators, but by theory the option value depends on only one of them, i.e. the volatility of the underlying contract during the option period. Since the case in hand is a future period, this indicator is unknown to the trader. However, if he wants to determine the theoretical value of an option, he will have to make a forecast of the underlying contract volatility for the option period [2].

It is quite difficult to exactly predict volatility, especially for a novice option trader. Traders are usually taught to predict price trend using technical analysis, and the information needed to do so can be picked up from multiple sources. In terms of volatility, there are much fewer materials, unfortunately [2].

Summarizing the above, it can be said that in order to predict the course of change in volatility, appropriate tools are needed. The volatility cone is one such tool.

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Model of the underlying contract price movement

For definiteness, we will talk about options on futures and denote the current futures price by the symbol F, but F can be understood as the current price of any underlying contract. It is assumed that the price dynamics of the underlying contract during a trading session is described by some continuous random process, and that there are no price jumps between sessions. Without getting into the mathematical details associated with correct representation of continuous random processes, we shall accept a simpler and clearer description of the price as a discrete process with some time step τ: $F_0 = F, F_1, F_2, \dots, F_i$. A step can be one day, one week, one hour, one minute, etc. The step will be expressed in fractions of a year, and since the process "exists" only during trading sessions, 1 year is considered equal to an average of 252 business days, and if, for instance, the time step is equal to one day (benchmark case), then [3]

$$\tau = \frac{1}{252}$$

Discrete-time geometric Brownian motion model:

$$\frac{\Delta F}{F_{i-1}} = \mu\tau + \sigma\sqrt{TZ_k}, \quad (11)$$

where

$\frac{\Delta F}{F_{i-1}} = \frac{F_i - F_{i-1}}{F_{i-1}}$ – relative change in price over the period τ;

μ – the average speed of the price trend, expressed as a simple annual interest rate;

$\sigma\sqrt{TZ_k}$ – irregular component:

σ – volatility;

$Z_1, Z_2, \dots, Z_i$ – independent Gaussian random variables with mathematical expectation equal to 0 and standard deviation equal to 1 (derived from normal distribution).

Volatility computing methods

There are different computational methods for historical volatility, such as close-to-close[11], Parkinson[12], Garman-Klass[13], Rogers-Satchell[14], Yang-Zhang extension for Garman-Klass, Yang-Zhang[15].

For definiteness, let's consider the close-to-close method. Historical volatility is calculated using the price of the underlying contract. The "window" principle is used for calculation, i.e. a calculation window groups a data stream according to the number of elements. A sliding calculation window slides over a data stream and counts the sum of the most recent elements every n elements.

In order to obtain an estimate of parameters μ and σ for a discrete set of prices of underlying contract $F_0 = F, F_1, F_2, \dots, F_i$, it is necessary to calculate:

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1. logarithmic change in price

   $u_i = \ln\left(\frac{F_i}{F_{i-1}}\right)$

   where $(i = 1, 2, \dots, n)$,

   u_i – continuously accrued return for the period τ,

   F_i – futures price at the end of the i-th interval$(i = 0, 1, 2, \dots, n)$.

   Value u_i can be defined as a relative change in price if

   $\left|\frac{F_i - F_{i-1}}{F_{i-1}}\right| \ll 1,$

   then the following approximation is done

   $$\ln\left(\frac{F_i}{F_{i-1}}\right) = \int_{F_{i-1}}^{F_i} \frac{dF}{F} \approx \int_{F_{i-1}}^{F_i} \frac{dF}{F_{i-1}} = \frac{F_i - F_{i-1}}{F_{i-1}}, \quad (12)$$

2. unbiased sample variance of σ_n^2 of value u_i for recent m measurements

   $$\sigma_n^2 = \frac{1}{m-1}\sum_{i=1}^{m}(u_{n-i} - \bar{u})^2, \quad (13)$$

   It is worth pointing out that formula (13) can be corrected by setting ū equal to zero, and having replaced denominator $m - 1$ with number m. These modifications have little effect on the variance estimates, but make it possible to rewrite formula (13) in a simplified form [4]:

   $$\sigma_n^2 = \frac{1}{m}\sum_{i=1}^{m} u_{n-i}^2, \quad (14)$$

   where values u_i are calculated using formula (12)

   In formula (14) the weights of all values are the same. Let's rewrite the formula, distributing greater weights to the recent data

   $$\sigma_n^2 = \sum_{i=1}^{m} \alpha_i \, u_{n-i}^2, \sum_{i=1}^{m} \alpha_i = 1 \quad (15)$$

   It is worth noting that the Fishburne method is used to weigh the elements [5]:

   $$\alpha_i = \frac{2(n-i+1)}{\big(n(n+1)\big)}, \quad (16)$$

   where

   α_i: weight coefficient of the i-th criterion, where $i = (1, 2, \dots, n)$;

   n: number of criteria in complete ordered series of criteria;

   i: number of the criterion in the complete complete ordered series of criteria.

3. average trend speed ū (mathematical expectation of the indicator u_i)

   $$\bar{u} = \frac{1}{m}\sum_{i=1}^{m} u_{n-i}, \quad (17)$$

4. volatility estimate for the period τ (grade s of standard deviation of values u_i)

   $$s = \sqrt{\sigma_n^2}, \quad (18)$$

Estimates of coefficients μ and σ are obtained by standardization:

$$\mu = \frac{\bar{u}}{\tau}, \sigma = \frac{s}{\sqrt{\tau}}, \quad (19)(20)$$

Calculation results and analysis

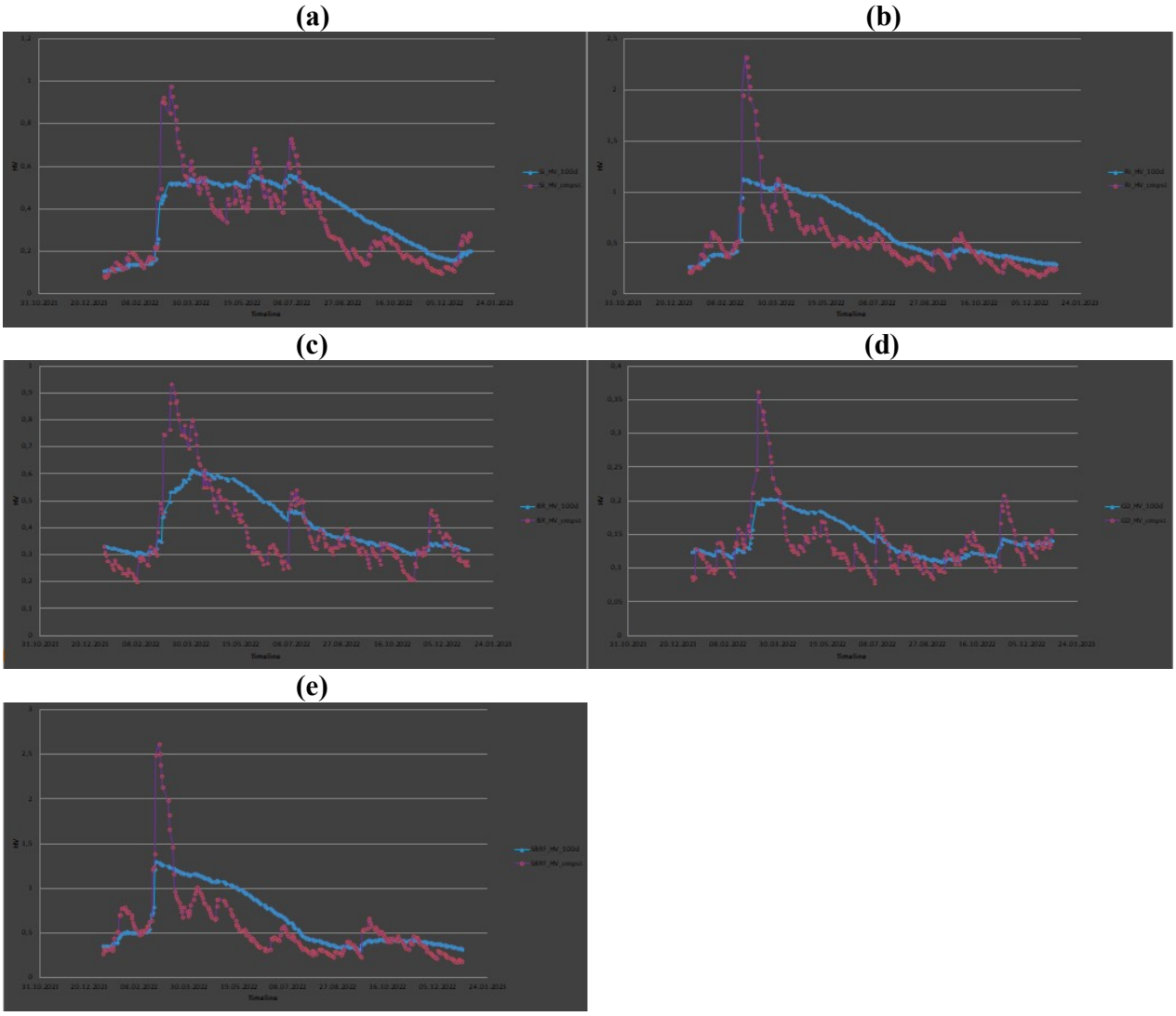


Figure 1. Curves of historical volatility calculated using the close-to-close method, where (a) is volatility of a US dollar-Russian ruble futures contract in 2022, (b) is volatility of a RTS Index futures contract in 2022, (c) is volatility of a Brent oil futures contract in 2022, (d) is volatility of a gold futures contract in 2022, (e) is volatility of a Sberbank PJSC ordinary shares futures contract in 2022. Volatility values in fractions are shown on the vertical axis to the left of the curves. Values in days are shown on the horizontal axis at the bottom of the figures. 100-day historical volatility points are marked in pink. Composite historical volatility points are marked in blue (consisting of the sum of 10-day with a weight of 0.4, 20-day with a weight of 0.3, 50-day with a weight of 0.2, 100-day with a weight of 0.1).

Forecasting volatility requires, first of all, an understanding of a number of its basic properties [2]. Based on curves (a – d), let us determine the main properties of volatility: changes in volatility are subject to trends, the underlying contract has long-term average volatility, and the volatility of the underlying contract is characterized by a return to the mean value.

Summarizing the above, it can be said that this is not enough to predict the course of change in volatility and evaluate options. Some constant value will be the result of historical volatility calculation. An option has a lifetime (time to expiration), and as a consequence is affected by time compression, i.e. the lifetime is constantly decreasing from the moment an option is launched until it expires. Volatility is a key parameter for option theoretical value computation. Given that $s = \sigma\sqrt{\tau}$, then volatility increases/decreases under the influence of time compression/expansion even when it is constant. Volatility should not simply be dynamic (GARCH(p,q) is a generalized autoregressive conditional heteroscedasticity model[16]), but it should be constantly recalculated taking into account constant time compression.

Volatility cone computation algorithm

1. Data preparation (annual distributions of futures contract prices). Quarterly futures require pre-adjustment, i.e., elimination of price discrepancies at contract renewal by combining selected fields from expired and continuing contract prices;
2. Computation of volatility estimates. For definiteness, the close-to-close method is used. Estimates are computed for several periods: 10-day, 20-day, 50-day, 100-day. The computation depth is 2 years (benchmark case). The depth is computed from the current date of the current year back to the beginning of the previous year;
3. Volatility ranking: all series of volatility estimates are sorted in ascending order; each series is divided by nine points into ten parts (deciles), each of which contains 10% of observations; in the first decile the minimum is determined, in all other deciles the maximum is determined;
4. Volatility cone representation. It is worth noting that the cone can be represented both tabularly and graphically. If the representation is tabular,
   then, according to point three, the ranked implied volatility series shall be added to the historical volatility series; color marks shall be added to indicate in which decile the current value of historical and implied volatility is located. If the representation is graphical, then the points of the upper boundary of the cone are plotted on the chart: 10-day maximum of the tenth decile, 20-day maximum of the tenth decile, 50-day maximum of the tenth decile, 100-day maximum of the tenth decile; points of the lower boundary of the cone: 10-day minimum of the first decile, 20-day minimum of the first decile, 50-day minimum of the first decile, 100-day minimum of the first decile; points of the center line of the cone: 10-day maximum of the fifth decile, 20-day maximum of the fifth decile, 50-day maximum of the fifth decile, 100-day maximum of the fifth decile. The points of each line are connected to each other (quantile lines can be plotted on the same principle). Additionally, points of current historical and current implied (from the central strike) volatility are plotted corresponding to a certain period, which depends on the expiration date of the options. As the time to expiration is constantly decreasing, the current historical volatility is constantly recalculated and a new point is plotted on the chart. It is also worth noting that updating the lows / highs of the corresponding periods of the lower/upper border of the cone is a trigger for redrawing.

[16] GARCH(p,q) is based on a formula in which variance value is determined from last p observations of the most recent relative change in price and last q estimates of the variance. GARCH(1,1) model was put forward by Bollerslev in 1986. Digits (1,1) in GARCH(1,1) model name mean that the most recent relative change in price and the most recent variance estimate are taken into account when computing the variance. GARCH(1,1) model is much more recognized than its other modifications.

Calculation results and analysis

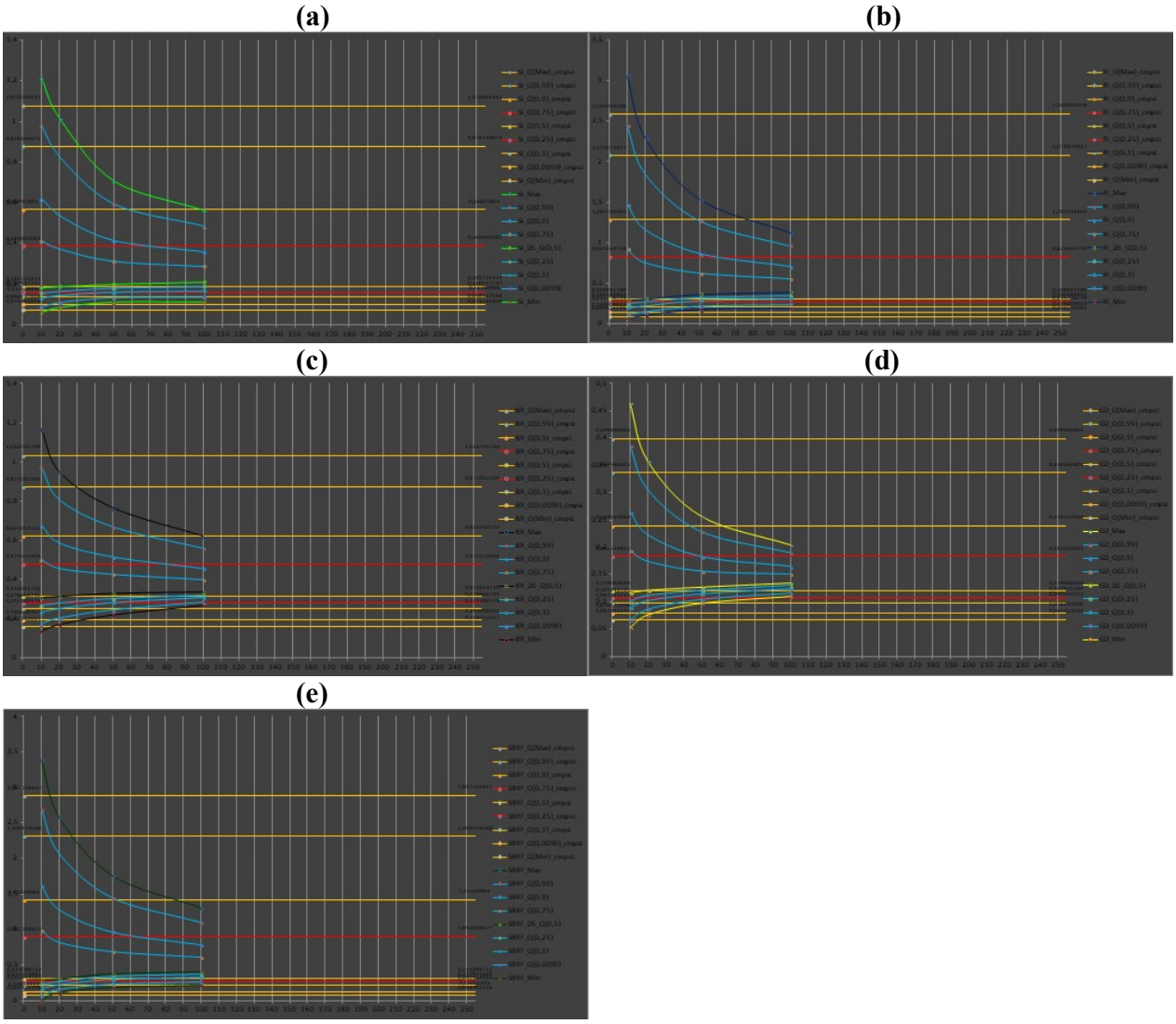


Figure 2. Curves of historical volatility calculated using the close-to-close method, where (a) is the volatility cone of a US dollar-Russian ruble futures contract in 2022, (b) is the volatility cone of a RTS Index futures contract in 2022, (c) is the volatility cone of a Brent oil futures contract in 2022, (d) is the volatility cone of a gold futures contract in 2022, (e) is the volatility cone of a Sberbank PJSC ordinary shares futures contract in 2022. Volatility values in fractions are shown on the vertical axis to the left of the curves. Values in days are shown on the horizontal axis at the bottom of the figures (number of days until expiration). Quantile lines are marked in blue.

Based on the graphs (a - d), it can be concluded that time compression/expansion is the main driver of expansion/contraction (volatility of the underlying contract converges to the long-term average) of the volatility cone.

Summarizing the above, it can be said that volatility should be forecast in the context of a cone; forecasting should be done according to a continuously decreasing period (use of volatility with a constant period will cause all volatility changes to be concentrated in one part of the cone); volatility should be continuously recalculated to account for constant time compression; assuming

that implied volatility follows historical volatility, comparing these volatilities in the context of a cone seems reasonable when making volatility decisions.

Conclusion

When trading options, every trader has a question of whether options are cheap or expensive. To answer this question, it is necessary to determine the value of the option. Volatility is a key parameter for option theoretical value computation. Hence, efforts spent on volatility forecasting by all available means can answer the question raised. The volatility cone is among such means.

All analytical computations and graphical constructions were performed using Microsoft Office Excel 2007 (starting from this version, several processor cores can be used to recalculate formulas, which increases the efficiency of calculations) with the use of VBA (Visual Basic for Applications) and are available in the public repository [6]. VBA is considered the standard scripting language for Microsoft applications. Thanks to VBA, we can create macros that greatly enhance Excel capabilities [7].

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